FolioQ Systems: Enterprise Cloud & SLA Architecture Report

Author: Engineering Architecture Team | Scope: FY2024 Global Infrastructure

1. Executive Overview and Architecture Strategy

The FolioQ distributed data platform utilizes a multi-region deployment across three active zones: North America (us-east-1, us-west-2), Europe (eu-west-1), and Asia-Pacific (ap-southeast-1). Each region maintains an independent storage tier with cross-region eventual consistency guaranteed within a 250ms replication window. During FY2024, the platform processed 14.8 billion transactions with a sustained average latency of 14.2 milliseconds across all read operations.

2. Service Level Agreements (SLA) & Incident Response Policies

Our Enterprise Tier SLA commits to 99.99% monthly availability for high-priority endpoints. Failure to maintain availability guarantees service credits according to the following policy:
- 99.90% to 99.98% availability: 10% monthly billing credit
- 99.00% to 99.89% availability: 25% monthly billing credit
- Below 99.00% availability: 50% monthly billing credit

Incident classification defines four operational severity levels:
• Severity 1 (Critical): Core service outage impacting >20% of active sessions. Response SLA: 15 minutes.
• Severity 2 (Major): Degradation of primary ingest or search pipeline. Response SLA: 45 minutes.
• Severity 3 (Moderate): Non-critical feature degradation with workaround. Response SLA: 4 hours.
• Severity 4 (Minor): Cosmetic UI defects or documentation updates. Response SLA: 24 hours.

3. Data Retention and Compliance Standards

All enterprise customer documents and extracted chunks are encrypted at rest using AES-256-GCM. Encryption keys are managed via AWS KMS with automatic annual key rotation. Backup snapshots are taken every 6 hours and stored across immutable WORM-compliant storage buckets. Audit logs are retained for a minimum of 7 years to comply with SOC 2 Type II, ISO/IEC 27001:2022, and HIPAA compliance requirements. Customer data deletion requests through GDPR Chapter III are executed within 72 hours, resulting in complete cryptographic erasure of all stored chunks.

4. Backup Recovery Time Objective (RTO) and Point Objective (RPO)

The formal disaster recovery specifications established for FY2024 are:
- Recovery Point Objective (RPO): Maximum 15 minutes of transactional data loss.
- Recovery Time Objective (RTO): Service restoration within 120 minutes in secondary regions.
- Annual disaster recovery simulation conducted on October 14, 2024 achieved failover in 47 minutes.